

MUHAMMAD UZAIR AFRIDI

AI ENGINEER

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PROFESSIONAL SUMMARY

AI Engineer with 5 years of experience owning AI solutions end to end, data ingestion and transformation pipelines, model development and evaluation, backend services and APIs, user-facing interfaces, deployment, and ongoing operations. Delivered production systems across both public sector (Saudi Ministry of Health) and private sector (Tier-1 downstream oil & gas) environments. Specialist in on-premise and air-gapped LLM infrastructure: GPU, RAM and storage provisioning with VRAM optimization on NVIDIA H100 hardware, local inference serving with Ollama and vLLM, containerized deployment with Docker and rootless Podman, and secure internal network configuration for internal AI services. Strong record of driving client consultation, presenting architecture options and trade-offs to stakeholders and translating operational requirements into deployed, compliant AI systems.

TECHNICAL SKILLS

Programming & Backend: Python, FastAPI, Django, Flask, Bash, SQL, REST API design, Webhooks, Enterprise integration interfaces

ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, XGBoost, Random Forest, Ensemble Models, Hyperparameter Tuning, Feature Engineering, Cross-Validation, Time-Series Forecasting (ARIMA/Prophet)

Generative AI & LLM Deployment: Ollama, vLLM, Open WebUI, LangChain, LangGraph, Prompt Engineering, PEFT/LoRA/QLoRA, LLaMA, Mistral, Gemma, Qwen, Azure OpenAI, Gemini, Structured model evaluation & benchmarking

Inference & Serving Optimization: KV-cache tuning, VRAM partitioning, quantization, batching & concurrency tuning, latency/throughput benchmarking, multi-model placement on shared GPUs

RAG & Retrieval Systems: RAG, Agentic RAG, Graph RAG, RAFT, Chunking Strategies, Embedding Models (BGE-M3), Cross-encoder Re-ranking, Hybrid Search, Vector Indexing

Infrastructure & MLOps: Docker, Podman (rootless), MLflow, DVC, CI/CD, Linux/RHEL/Debian administration, SELinux, GPU provisioning & VRAM optimization (NVIDIA H100 NVL), Air-gapped deployment, Model monitoring, retraining & drift tracking

Cloud & Data Platforms: AWS (ML microservice deployment), Azure (Azure OpenAI, agentic workloads), on-premise private cloud architecture

Data Engineering: Ingestion, transformation and processing pipelines for structured and unstructured data, Pandas, NumPy, Data Validation, EDA, Statistical Analysis

Databases & Vector Stores: PostgreSQL, MySQL, Pinecone, ChromaDB, Qdrant, NoSQL, JSON

Enterprise Integration: Identity and access management / role-based access gating, authentication services, database and enterprise application integration, third-party API and service integration

Security & Compliance: Prompt-injection filtering, PII redaction, LLM proxy guardrails, Data residency & sovereignty controls, Red-team testing of LLM stacks, Security architecture documentation

Interfaces & Frontend: JavaScript, HTML, CSS, Streamlit, Open WebUI customization (Filters, Tools, Pipe functions), Operational dashboards

AI-Assisted Development: Claude Code, Cursor, GitHub Copilot — used to accelerate prototyping, refactoring and delivery of production services

PROFESSIONAL EXPERIENCE

AI Specialist | AI Sawab Commercial Company

Yanbu, KSA | 06/2026 – Present

- Own end-to-end delivery of an AI contract and procurement intelligence platform for a Tier-1 Saudi downstream oil & gas client — covering document ingestion pipelines, retrieval architecture, FastAPI backend services, the Open WebUI interface layer, on-premise deployment, and ongoing operations.
- Designed and provisioned the on-premise inference infrastructure on dual NVIDIA H100 NVL GPUs — sizing GPU, RAM and storage, then tuning vLLM KV-cache allocation, VRAM partitioning and model placement to serve multiple concurrent workloads without contention.
- Optimized inference performance and latency under concurrent enterprise load through batching, context-length and quantization trade-off testing, benchmarking throughput and time-to-first-token to hit agreed response-time targets.
- Built production data ingestion and transformation pipelines over unstructured enterprise governance and contract corpora — parsing, chunking, BGE-M3 embedding and vector indexing with cross-encoder re-ranking — plus a deterministic Python comparison engine for invoice rate-compliance validation against contracted rate sheets.
- Developed backend services and REST APIs exposing AI capabilities securely to internal consumers, and integrated the platform with client enterprise systems, databases and role-based access controls.
- Built and customized the user-facing interface and operational dashboards (Open WebUI Filters, Tools and Pipe functions) so business users could run contract drafting, review and compliance workflows without engineering support.
- Engineered an LLM security gateway acting as a reverse proxy in front of locally hosted models — enforcing prompt-injection filtering, PII redaction and identity-based access gating — and architected secure internal network configuration for air-gapped operation, eliminating outbound dependencies, isolating inference service bindings and hardening the container surface using rootless Podman on RHEL with SELinux.
- Ran structured model evaluations across candidate open-weight LLMs against defined production use cases (contract drafting, contract review, invoice compliance), documenting accuracy, latency and safety-filter behaviour to drive the selection decision.
- Monitor deployed models in production — tracking output quality, retrieval accuracy and failure modes — and iterate on retrieval configuration, prompts and model selection based on observed drift and user feedback.
- Conducted red-team and prompt-injection testing of the deployed stack and produced the security architecture documentation for the production environment.

- Drove client-facing consultation, presenting architecture options, trade-offs and delivery roadmaps directly to stakeholders and translating operational requirements into implemented AI capabilities.

AI Engineer (Public Sector Contract) | Tech Track — Ministry of Health Hajj Project

Riyadh, KSA | 03/2026 – 05/2026

- Delivered a RAG-powered case report generation system for the Ministry of Health's Hajj deceased management workflow, automating structured documentation (cause of death, repatriation procedures, next-of-kin notification) grounded against MoH medical protocols and regulatory guidelines.
- Designed the full data ingestion and processing pipeline for unstructured medical and procedural documents — parsing, chunking, embedding, vector indexing and retrieval strategy tuning for factual accuracy in a high-stakes regulatory context.
- Deployed locally hosted open-source LLMs to satisfy government data residency and sovereignty requirements, with zero external API dependencies.
- Containerized the full backend and frontend stack with Docker for air-gapped deployment on on-premise Debian servers, and handed over operational documentation and runbooks to the client team.

AI Engineer | JMM Technologies

Peshawar, Pakistan | 06/2024 – 01/2026

- Designed and deployed embedding-based retrieval and RAG pipelines using LangChain, LangGraph and Pinecone, improving enterprise knowledge retrieval accuracy by 70% through chunking, embedding and re-ranking optimization.
- Scaled AI capabilities across 3+ enterprise workflows by containerizing models with Docker and exposing them through FastAPI/REST microservices, with monitoring and alerting for model performance, data drift and service reliability.
- Built agentic AI systems with multi-step query understanding and intent detection, running autonomous reasoning across enterprise data sources in production on Azure and AWS cloud ML environments.
- Delivered ~2.1% RMSE on industrial time-series forecasting using ensemble models (Random Forest, XGBoost) with hyperparameter tuning, uncertainty quantification and cross-validated model selection.
- Integrated open-source LLMs (Mistral, LLaMA) and hosted APIs (Azure OpenAI, Gemini) with prompt engineering and PEFT fine-tuning for domain-specific retrieval and generation tasks.

Machine Learning Engineer | DevMaze

Peshawar, Pakistan | 06/2023 – 05/2024

- Built semantic search and RAG systems handling embedding, indexing, query routing and retrieval over unstructured text, reducing retrieval errors and improving end-user answer accuracy.
- Developed and fine-tuned Transformer-based NLP pipelines with Hugging Face and PyTorch for text classification, entity extraction, question answering and summarization.
- Improved model performance by 25% on regression and classification tasks through systematic feature engineering, cross-validation and comparative evaluation across model variants.
- Established reproducible experimentation workflows with versioned data pipelines and structured experiment tracking, reducing rework across model iterations.

Junior Python Developer | Ray Media Group

Peshawar, Pakistan | 06/2021 – 06/2023

- Built modular, PEP8-compliant Python backend services and RESTful API integrations powering automated analytics pipelines.
- Optimized complex SQL queries on PostgreSQL and MySQL and designed efficient schemas, improving data reliability for downstream modelling and analysis.
- Translated business requirements from content, marketing and operations teams into delivered technical implementations.

SELECTED PROJECTS

- **Agentic AI Research Assistant** — Fully local agentic system on Ollama and Mistral with MCP-style tool orchestration and real-time web search, demonstrating end-to-end query understanding, retrieval orchestration and ranked result delivery.
- **Smart Finance Assistant** — Full-stack AI application with a FastAPI microservice backend, Streamlit frontend and a local LLM for real-time data processing and personalized recommendations.
- **Sales Demand Forecasting** — End-to-end time-series ML pipeline (ARIMA/Prophet plus tree-based models) with cross-validated model comparison, translating model outputs into business decision scenarios.

EDUCATION

Bachelor of Science in Software Engineering — IMSciences University, Peshawar, Pakistan

CERTIFICATIONS

DeepLearning.AI: Supervised Machine Learning • Introduction to TensorFlow for AI • NLP with Classification and Vector Spaces | Dice Analytics: Python for Machine Learning | IBM: Introduction to Data Science, Introduction to AI